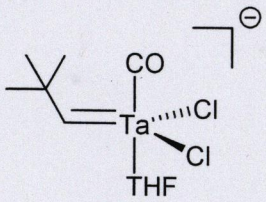
Inorganic and Materials Chemistry
Cumulative Exam: 9-8-2017

Instructions: Answer any five of the following ten questions. Only <u>your first five responses will be graded</u> the rest will be ignored. each question is worth 10 pts.

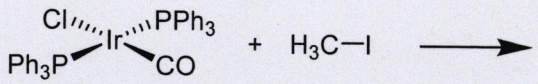
Solution state inorganic chemistry

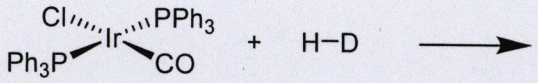
Using representation theory and the attached character tables, answer the following questions for SbF_5^{2-}

1. What is the point group of SbF_5^{2-} ?
2. Provide a reducible representation for all molecular motions (Γ_{3N}) for SbF_5^{2-}.
3. Using group theory determine the total number of IR and Raman active vibrational bands for SbF_5^{2-} (show all of your work.)
4. What is the oxidation state of the metal, number of d electrons (d count), and total number of valence electrons in the complex for the structure drawn below?



5. Predict the products of the following two reactions and provide a complete mechanism for each reaction.





Solid state and materials chemistry.

6. Sketch the unit cell for basic cubic, face centered cubic and body centered cubic crystal structures of pure metals.
7. Describe a p doped semiconductor, include a description of the band structure and the fermi level relative to the undoped material. (you may use any diagrams or sketches if you think they will be helpful).
8. Describe an n doped semiconductor, include a description of the band structure and the fermi level relative to the undoped material. (you may use any diagrams or sketches if you think they will be helpful).
9. Describe the experimental evidence for superconductivity.
10. Define the Tafel law. What experimental evidence / observations would you expect for an electrochemical reaction that obeys the Tafel law?

$$\#\,reps = \frac{1}{h}\sum(g * \chi_r * \chi_i)$$

h = total operations in a group

g = number of operations in a class

χ_r = the character of the reducible representation

χ_i= the character of the irreducible representation

The transformation matrix for a C_n rotation around the z axis is

$$\begin{bmatrix} cos\,\theta & -sin\theta & 0 \\ sin\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$cos\frac{2\pi}{1} = 1, sin\frac{2\pi}{1} = 0$

$cos\frac{2\pi}{2} = -1, sin\frac{2\pi}{2} = 0$

$cos\frac{2\pi}{3} = -0.5, sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

$cos\frac{2\pi}{4} = 0, sin\frac{2\pi}{4} = 1$

$cos\frac{2\pi}{5} = \frac{1}{4}(-1+\sqrt{5}), sin\frac{2\pi}{5} = \frac{1}{4}\sqrt{10+2\sqrt{5}}$

$cos\frac{2\pi}{6} = 0.5, sin\frac{2\pi}{6} = \frac{\sqrt{3}}{2}$

Character Tables

C_2	E	C_2		
A	1	1	z, R_z	x^2, y^2, z^2, xy
B	1	-1	x, y, R_x, R_y	yz, xz

C_3	E	C_3	$(C_3)^2$		
A	1	1	1	z, R_z	x^2+y^2, z^2
E	1 1	e e^*	e^* e	x+iy; R_x+iR_y x-iy; R_x-iR_y	(x^2-y^2, xy) (yz, xz)

e = exp(2πi/3)

C_4	E	C_4	C_2	$(C_4)^3$		
A	1	1	1	1	z, Rz	x^2+y^2, z^2
B	1	-1	1	-1		x^2-y^2, xy
E	1 1	i -i	-1 -1	-i i	x+iy; R_x+iR_y x-iy; R_x-iR_y	(yz, xz)

C_{2v}	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	z	x^2, y^2, z^2
A_2	1	1	-1	-1	R_z	xy
B_1	1	-1	1	-1	x, R_y	xz
B_2	1	-1	-1	1	y, R_x	yz

C_{3v}	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	-1	R_z	
E	2	-1	0	(x, y) (R_x, R_y)	(x^2-y^2, xy) (xz, yz)

C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$		
A_1	1	1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z	
B_1	1	-1	1	1	-1		x^2-y^2
B_2	1	-1	1	-1	1		xy
E	2	0	-2	0	0	(x, y) (R_x, R_y)	(xz, yz)

C_{6v}	E	$2C_6$	$2C_3$	C_2	$3\sigma_v$	$3\sigma_d$		
A_1	1	1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	1	-1	-1	Rz	
B_1	1	-1	1	-1	1	-1		
B_2	1	-1	1	-1	-1	1		
E_1	2	1	-1	-2	0	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	=1	=1	2	0	0		$(x^2-y^2$, xy)

C_{5v}	E	$2C_5$	$2(C_5)^2$	$5\sigma_v$		
A_1	1	1	1	1	z	x2+y2, z2
A_2	1	1	1	-1	Rz	
E_1	2	2cos(2π/5)	2cos(4π/5)	0	(x, y) (Rx, Ry)	(xz, yz)
E_2	2	2cos(4π/5)	2cos(2π/5)	0		(x2-y2, xy)

D_{2h}	E	C_2 (z)	C_2 (y)	C_2 (x)	i	σ (xy)	σ (xz)	σ (yz)		
A_g	1	1	1	1	1	1	1	1		x^2, y^2, z^2
B_{1g}	1	1	-1	-1	1	1	-1	-1	R_z	xy
B_{2g}	1	-1	1	-1	1	-1	1	-1	R_y	xz
B_{3g}	1	-1	-1	1	1	-1	-1	1	R_x	yz
A_u	1	1	1	1	-1	-1	-1	-1		
B_{1u}	1	1	-1	-1	-1	-1	1	1	z	
B_{2u}	1	-1	1	-1	-1	1	-1	1	y	
B_{3u}	1	-1	-1	1	-1	1	1	-1	x	

D_{3h}	E	$2C_3$	$3C'_2$	σ_h	$2S_3$	$3\sigma_v$		
A'_1	1	1	1	1	1	1		x^2+y^2, z^2
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	(x^2-y^2, xy)
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

D_{4h}	E	$2C_4$	C_2	$2C'_2$	$2C''_2$	i	$2S_4$	σ_h	$2\sigma_v$	$2\sigma_d$		
A_{1g}	1	1	1	1	1	1	1	1	1	1		x^2+y^2, z^2
A_{2g}	1	1	1	-1	-1	1	1	1	-1	-1	Rz	
B_{1g}	1	-1	1	1	-1	1	-1	1	1	-1		x^2-y^2
B_{2g}	1	-1	1	-1	1	1	-1	1	-1	1		xy
E_g	2	0	-2	0	0	2	0	-2	0	0	(Rx, Ry)	(xz, yz)
A_{1u}	1	1	1	1	1	-1	-1	-1	-1	-1		
A_{2u}	1	1	1	-1	-1	-1	-1	-1	1	1	z	
B_{1u}	1	-1	1	1	-1	-1	1	-1	-1	1		
B_{2u}	1	-1	1	-1	1	-1	1	-1	1	-1		
E_u	2	0	-2	0	0	-2	0	2	0	0	(x, y)	

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$		
A_1	1	1	1	1	1		$x^2+y^2+z^2$
A_2	1	1	1	-1	-1		
E	2	-1	2	0	0		$(2z^2-x^2-y^2, x^2-y^2)$
T_1	3	0	-1	1	-1	(Rx, Ry, Rz)	
T_2	3	0	-1	-1	1	(x, y, z)	(xy, xz, yz)